EXERCÍCIOS NÚMEROS COMPLEXOS

$x_1 = 2 + j3$, $x_2 = 4 - j3$, $x_3 = 5 \angle 0{,}927$ $\rightarrow x_3 = 3 + 4i$

$\hookrightarrow a = 5 \cdot \cos(0{,}927)$

$a \cong 3$

$b = 5 \cdot \operatorname{sen}(0{,}927)$

$b = 4$

a) $x_1 + x_2$

$2 + \cancel{3j} + 4 - \cancel{3j}$

6

b) $x_2 + x_3$

$4 - 3j + 3 + 4j$

$= 7 + j$

c) $x_2 - x_3$

$4 - 3j - (3 + 4j)$

$4 - 3j - 3 - 4j$

$1 - 7j$

D) $x_2 \cdot x_3$

$(4 - 3j) \cdot (3 + 4j)$

$12 + 16i - 9j - 12j^2$

$12 + 7j + 12$

$24 + 7j$

$j^2 = \sqrt{-1}^2 = -1$

e) x_2 / x_3

$$\frac{4 - 3j}{3 + 4j} \cdot \left(\frac{3 - 4j}{3 - 4j}\right)$$

$$\frac{\cancel{12} - 16j - 9j + \cancel{12j^2}}{9 - 16j^2} = -\frac{25j}{+25} = -j$$

$j^2 = -1$

f) $Re(x_3) = 3$

g) $Im(x_3) = 4j$

h) $|x_1| = \sqrt{a^2 + b^2}$

$a = 2$

$b = 3$

$= \sqrt{2^2 + 3^2}$

$= \sqrt{13}$

2) a) $(6 + 7j) \cdot (1 + j)$ $\quad j^2 = -1$

$= 6 + 6j + 7j + 7\cancel{j^2}^{-1}$

$= -1 + 13j$

b) $(5+j4)\cdot(1-j)+(2+j)j$

$= 5-5j+4j-4j^2+2j+j^2$

$= 5+1j-3j^2$

$= 8+j$

c) $(1+2j)^2-(3+4j)$

$(a+b)^2 = (a+b)\cdot(a+b)$

$= a^2+2ab+b^2$

$1+\cancel{4j}+4j^2-3-\cancel{4j}$

-6

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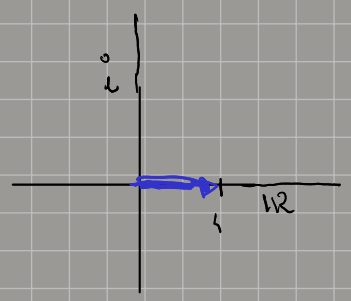
3) $|z| = \sqrt{a^2+b^2}$, $\theta = \text{ARC}tg\left(\frac{b}{a}\right)$

a) 4

$|z| = \sqrt{4^2+0} = 4$

$\theta = tg^{-1}\left(\frac{0}{4}\right)$

$z = 4\angle 0^\circ$

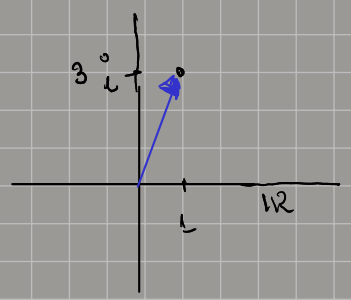


b) $1+\sqrt{3}j$

$|z| = \sqrt{1^2+\sqrt{3}^2} = \sqrt{1+3} = 2$

$\theta = tg^{-1}\left(\frac{\sqrt{3}}{1}\right) = 60^\circ$

$z = 2\angle 60^\circ$

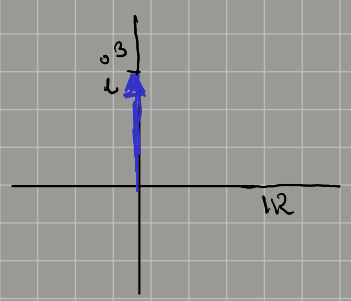


c) $3j$

$|z| = \sqrt{3^2} = 3$

$\theta = tg^{-1}\left(\frac{3}{0}\right) = 90^\circ$

$z = 3\angle 90^\circ$



d) $-\sqrt{2}+\sqrt{2}j$

$|z| = \sqrt{(-\sqrt{2})^2+\sqrt{2}^2}$

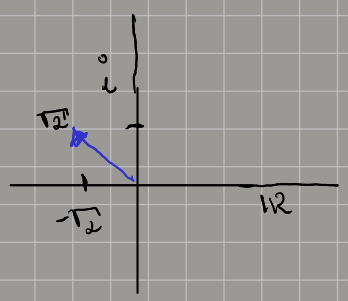
$|z| = \sqrt{2+2}$

$= \sqrt{4}$

$= 2$

$\theta = tg^{-1}\left(\frac{\sqrt{2}}{-\sqrt{2}}\right)$

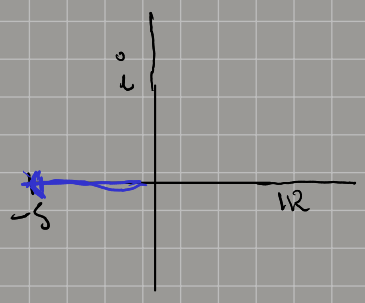
$\theta = -0{,}785$



e) -5

$|z| = \sqrt{(-5)^2 + 0^2} = \sqrt{25} = 5$

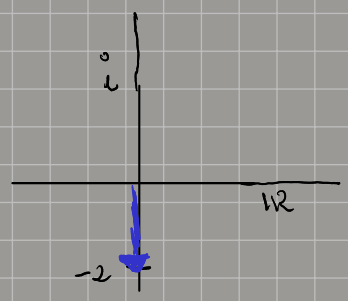
$\theta = tg^{-1}\left(\frac{0}{-5}\right) = 180^\circ$.



f) $-2j$

$|z| = \sqrt{0^2 + (-2)^2} = \sqrt{4} = 2$

$\theta = tg^{-1}\left(\frac{-2}{0}\right) = 270^\circ$



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4) a) $3 - 4j$

$|z| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$\theta = tg^{-1}\left(\frac{-4}{3}\right) = -53{,}13^\circ$

b) $\sqrt{2} + \sqrt{2}j$

$|z| = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{4} = 2$

$\theta = tg^{-1}\left(\frac{\sqrt{2}}{\sqrt{2}}\right) = 45^\circ$

c) $12 + 5j$

$|z| = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = \sqrt{169} = 13$

$\theta = tg^{-1}\left(\frac{5}{12}\right) = 22{,}62^\circ$ $\quad z = 13 < 0{,}30{,}5^\circ$

d) $\cos(\theta) + j\,sen(\theta)$

$|z| = \sqrt{\cos^2 + sen^2} = \sqrt{1} = 1$

$\theta = \theta$

e) $-3+4j$

$$|z| = \sqrt{(-3)^2+4^2} = \sqrt{25} = 5$$

$$\theta = Tg^{-1}\left(\frac{4}{-3}\right) = -53{,}13^\circ$$ COMO ESTÁ NO 2° Q, $\theta = -53{,}13 + 180$

$\theta = 126{,}87^\circ$

f) $7-2j$

$$|z| = \sqrt{7^2+(-2)^2} = \sqrt{49+4} = \sqrt{53}$$

$$\theta = Tg^{-1}\left(\frac{-2}{7}\right) = -15{,}9^\circ$$

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